



The Aeronautical Society of India
Design Division & Mumbai Branch
present



9th National Aerospace Conceptual Design Competition (NACDeC-IX)

Tactical Hunting And Reconnaissance (THAR) UAV

1. Mission summary

Design an unmanned aerial vehicle (*THAR UAV*) to perform persistent surveillance along the Indian border in the Thar desert, without the need for a runway. *THAR UAV* should be capable of being launched and recovered from a 50 m x 50 m prepared area in a desert, and then conducting a border-following patrol. Its objective is to detect human activity in day and night conditions using an EOIR turret payload, and on confirmed detection, deliver a small non-lethal neutraliser to temporarily incapacitate or slow down the target until ground forces arrive at that location.

2. Special Considerations

2.1 Runway Independence

Since the *THAR UAV* is required to operate from sandy terrains in the deep desert, it should not require a runway for take-off or landing. Due to the operational difficulty of deployment of bulky equipment in such terrains, aircraft launchers and catapults shall not be used.

2.2 Transportability

The *THAR UAV* system should be fully dismantlable and portable in a single truck. A truck with a standard container (2 m x 3 m x 3 m) should be capable of carrying the aircraft in disassembled condition to the launch location.

2.3 Onboard Avionics

The avionics suite must include essential sensors like GPS/ GNSS/INS, communications datalink for Command and Control, Image Downlink, Flight Control Computer and Engine Control Unit with engine accessories. A proposed avionics architecture and weight budget must be presented.

3. CONOPS (Concept of Operations)

1. **Deploy:** Truck drives to remote location. Field crew assembles airframe and payloads
2. **Preflight:** Systems check, mission plan upload
3. **Launch:** Runway-independent launch and climb to cruise altitude of 3000 m
4. **Cruise Patrol:** Cruise @ 3000 m, follow preplanned border track, maintain surveillance. Transmit the image seen by the EOIR turret payload to the ground station at the launch location every one minute.
5. **Detect:** On detection of suspicious human movement, aircraft verifies target
6. **Engage:** Drop Neutraliser Capsules on verified target
7. **Loiter:** Continue Step 4 till all Neutraliser Capsules are released, or fuel remaining is just enough to return to deployment location. Transmit image seen by the
8. **Return & recover:** Return to deployment location, recover to truck, turn around

4. Patrolling Distance

The *THAR* UAV should be designed such that the aircraft can cover at least half of the length of the international border in the Thar desert before it turns around to return to the deployment location. The total length of the international border should be taken as the exact length of the contour of the border from Rankhar in the south to Sri Ganganagar in the north.

So, patrolling distance for *THAR* UAV should be considered either from Rankhar to Longewala and back, or from Sri Ganganagar to Longewala and back, as marked in Figure 1.



Figure 1 Map of THAR UAV Mission

The EOIR turret camera is equipped to operate at a stand-off distance of 3 km, safely within Indian airspace. Therefore, the aircraft is never required to cross the border, but in fact should stay well clear of it by a margin of 3 km.

5. Aircraft Specifications – THAR UAV

Table 1 THAR UAV Specifications

Specification	Value
Launch and Recovery	Runway Independent
	Prepared area 50 m x 50 m
Total flight duration from launch to recovery	> 9 hours
Cruise Altitude	3000 m AMSL
MTOW	< 175 kg
Payload weight	15 kg (EOIR Turret + Neutraliser)
Wing span	< 6 m
Sustained Turn	1.3 g
Operational Environment	-10 to 45 deg C 50% RH Dusty
Reference Atmosphere	Indian Reference Atmosphere (IRA)

6. Payload

THAR UAV should be designed to carry two distinct payloads to fulfil its mission, namely the EOIR Turret and the Neutraliser Capsule.

6.1 EOIR Turret

The EOIR Turret is a stabilized, gimballed sensor package (total mass ≈ 5 kg) providing day/night detection and classification for clandestine human activity. It combines a high-resolution Electro-Optical (EO) camera with an InfraRed (IR) detector, optical zoom and image stabilization. It can detect and recognise a human from aircraft cruise altitude at a depression angle of 45 degrees. The EOIR turret is to be mounted in the nose or belly of the aircraft with a clear field of view all around it. While the aircraft operates in fully autonomous mode, it is required that one image is to be transmitted every one minute back to the ground-based controller located at the launch location. To serve this purpose, an appropriate Beyond Visual Line of Sight (BVLoS) communication system like a small satellite link unit should be included as part of the avionics suite.

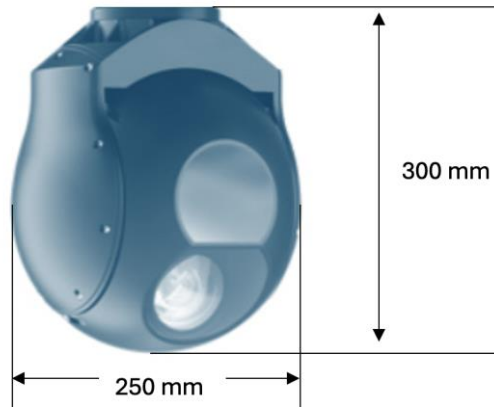


Figure 2 EOIR Turret Dimensions

Table 2 EOIR Turret Specifications

Parameter	Specification
Weight of EOIR Turret	5 kg
Location requirement	Clear field of view forward and downward
Azimuth	360 degrees
Elevation	0 to -90 degrees

6.2 Neutraliser Capsule

The neutraliser is a non-lethal device containing a disorienting but medically harmless marker, the effects of which are reversible and certified medically safe. It is optimised to fit in a capsule weighing 1 kg which should be dropped from the UAV from cruise altitude when it spots illegal movement on the ground. Four capsules are mounted under a launcher as shown in Figure 3. The launcher should be carried in the belly of the aircraft.

Table 3 Neutraliser Capsule Specifications

Parameter	Specification
Length of Neutraliser Capsule	200 mm
Diameter of Neutraliser Capsule	50 mm
Weight of Neutraliser Capsule	1 kg
Number of Neutraliser Capsules carried	4
Weight of Neutraliser Capsule Launcher	6 kg



Figure 3 Front view of Neutraliser Capsules with Launcher

7. Stagewise Deliverables

Stage 1 (8 weeks)	Initial Design Report (IDR) showing: 1. Literature Survey of Similar UAVs 2. Specifications Design Drivers of THAR 3. Basic Design Philosophy adopted for THAR 4. Selection of Baseline Configuration 5. Estimation of Aerodynamic Characteristics (using Empirical methods) 6. Initial Sizing & Constraint Analysis 7. Selection of Propulsion System 8. Basic Sizing of Stabilizing & Control Surfaces
Stage 2 (8 weeks)	Final Design Report (FDR) showing: 1. Executive Summary 2. Three View Diagram (CAD) & Inboard View 3. Weight breakdown 4. Propulsion System details 5. Avionics Architecture 6. Stability & Control Assessment 7. Materials Manufacturing Philosophy 8. Mission Performance Compliance Table

An aircraft that meets all requirements and has the lowest value of the following *Figure of Merit (FoM)* will be considered superior.

$$FoM = 0.5 * \frac{MTOW}{175} + 0.5 * \frac{Wingspan}{6.0}$$